i) $\nabla f|_{P_0} = 2xi + 2yj + 2zk(1,1,1) = 2i + 2j + 2k$

Plano Tang

$2(x-1) + 2(y-1) + 2(z-1) = 0 \Rightarrow 2x - 2 + 2y - 2 + 2z - 2 = 0$

$\Rightarrow 2x + 2y + 2z = 6 \Rightarrow x + y + z = 3$

Reta NORMAL:

$x = 1 + 2t \,,\; y = 1 + 2t \,,\; z = 1 + 2t$